

Clinical Risk Assessment Report: Patient 72

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 44.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.99x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +6.2%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.04 [Top 10%] (+6.2%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.41 [Bottom 2%] (+6.2%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 55.0 [Bottom 22%] (+4.5%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.99 [Top 14%] (+4.1%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.77 [Top 4%] (+3.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.66 [Avg (67th %ile)] (+2.7%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.24 [Bottom 6%] (+2.4%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.14 [Top 13%] (-2.8%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)

